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**Report #2**  
**Experiment No. 10**  
**Cisco ASA Firewall Configuration**

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May – 2025



## Abstract

This experiment focuses on configuring and securing a network using Cisco ASA Firewall in a simulated environment( packet tracer). The main objectives were to verify firewall functionalities including Network Address Translation (NAT), Access Control Lists (ACLs), DHCP, and interface configuration. We also used dynamic routing protocols-OSPF and BGP-with route redistribution to simulate communication between internal and external autonomous systems. Key deliverables included a functional network with secured DMZ access, internal wireless connectivity, and successful external access to hosted services via static NAT. Cisco Packet Tracer was used as the primary tool to simulate and test connectivity, security policies, and protocol behaviors. The expected result was to establish secure communication across the internal network, DMZ, and the Internet, and that was done through practical test cases.

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## Acronyms and Abbreviations

|             |                                            |
|-------------|--------------------------------------------|
| <i>OSPF</i> | <i>Open Shortest Path First</i>            |
| <i>BGP</i>  | <i>Border Gateway Protocol</i>             |
| <i>AS</i>   | <i>Autonomous System</i>                   |
| <i>IGP</i>  | <i>Interior Gateway Protocol</i>           |
| <i>EGP</i>  | <i>Exterior Gateway Protocol</i>           |
| <i>NAT</i>  | <i>Network Address Translation</i>         |
| <i>PAT</i>  | <i>Port Address Translation</i>            |
| <i>DMZ</i>  | <i>Demilitarized Zone</i>                  |
| <i>ASA</i>  | <i>Adaptive Security Appliance</i>         |
| <i>ACL</i>  | <i>Access Control List</i>                 |
| <i>NGFW</i> | <i>Next Generation Firewall</i>            |
| <i>VPN</i>  | <i>Virtual Private Network</i>             |
| <i>DHCP</i> | <i>Dynamic Host Configuration Protocol</i> |
| <i>RR</i>   | <i>Resource Records</i>                    |

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# 1 Introduction

A firewall is an integral part of network security that controls data communication among networks by filtering incoming and outgoing packets. Based on a pre-established set of rules-typically IP addresses, port numbers, and protocols-the firewall will accept or reject some traffic. [1]

Firewalls act as a barrier between an internal trusted network (e.g., an internal LAN) and the untrusted external networks (including the Internet). They exist to block unauthorized traffic and filter threats from malicious sources, i.e., hackers, malware, and bots. Without protection from this layer, internal systems would be vulnerable to data breaches and services loss. [1]

## 1.1 Firewalls: Forms and Implementations

1. Packet-Filtering Firewalls: Examine every packet on a packet-by-packet basis based on header information (e.g., IP address, port). [1]
2. Stateful Inspection Firewalls: Track active sessions and analyze the context of the traffic, if the packet were sent from “Inside” it will allow responding from “Outside”. [1]
3. Proxy Firewalls: Function as a mediator between users and the external network, hiding internal addresses. [1]
4. Next-Generation Firewalls (NGFWs): Enable advanced features such as deep packet examination, intrusion detection, and application-level filtering. [1]
5. Cloud Firewalls: Provide similar protection but reside in cloud infrastructures, which is well adapted to cloud-based infrastructure. [1]

## 1.2 Cisco ASA. Overview

Cisco ASA is a multi-purpose, high-speed security appliance that combines its several different security tools within its platform. It has supports following services

1. Stateful firewall. [1]
2. Intrusion prevention. [1]
3. Antivirus. [1]
4. Support for VPN (Virtual Private Network). [1]

In default configuration, the ASA blocks all incoming traffic from outside, hence isolating the internal network. But it has its flexible rules and inspections, which permit authorized external connections (e.g., looking through internal web or DNS servers) under strict control. [1]

### 1.3 Security Levels in Cisco ASA Firewalls

In Cisco ASA (Adaptive Security Appliance) firewalls, each network interface is assigned a security level between 0 and 100, which determines the trustworthiness of the connected network zone. This makes traffic control easier by implicitly establishing default actions between zones of varying trust.[1]

Typical Security Level Assignments:

1. Inside (Security Level 100): This is the most trusted zone, usually the internal LAN where the users, workstations, and sensitive data reside.[1]
2. DMZ (Security Level 50): The Demilitarized Zone is a partially trusted zone that is meant to be utilized for publicly accessible services like web servers, DNS, and FTP. It is isolated from both the internal LAN and the external Internet to form a security zone.[1]
3. Outside (Security Level 0): The least trusted zone, normally connected to the Internet or other untrusted networks. Traffic from here is presumed to be potentially dangerous.[1]

Figure 1-1 summarize the latter

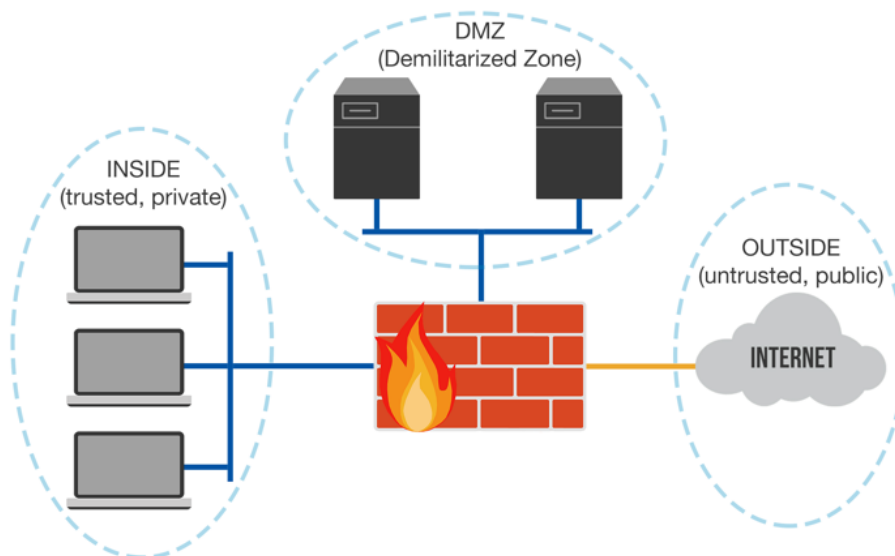


Figure 1-1: ASA illustration sketch. [2]

In a typical ASA flowing from a higher to a lower security level is allowed whereas flowing from lower security level to a higher security level is not allowed. In our experiment, to overcome this problem for specific traffic we will use Access Control Lists (ACL). [1]

#### 1.4 ASA capabilities: NAT (Network Address Translation), DHCP and Routing

NAT allows internal devices with private addresses to talk to outside networks. Cisco ASA has support for:

1. Static NAT: Translates one private address to one public address-perfect for servers that should be always reachable.[1]. It can be configured as follows:

```
ciscoasa(config)# object network <object-name>  
ciscoasa(config-network-object)# host <private-ip>  
ciscoasa(config-network-object)# nat (<in-if>, <out-if> ) static <public-ip>
```

2. Dynamic NAT: Translates private addresses to a group of public addresses-utilized when multiple internal hosts should occasionally access the Internet.[1]
3. PAT (Port Address Translation): Maps several private IPs to one public IP with different port numbers-efficient in conserving public IPs and is most widely used.[1] it can be configured using following commands:

```
ciscoasa(config)# object network <object-name>  
ciscoasa(config-network-object)# subnet <private-ip> <subnet mask>  
ciscoasa(config-network-object)# nat (<in-if>, <out-if> ) dynamic interface
```

Figure 1-2 shows a schematic sketch of how PAT is working

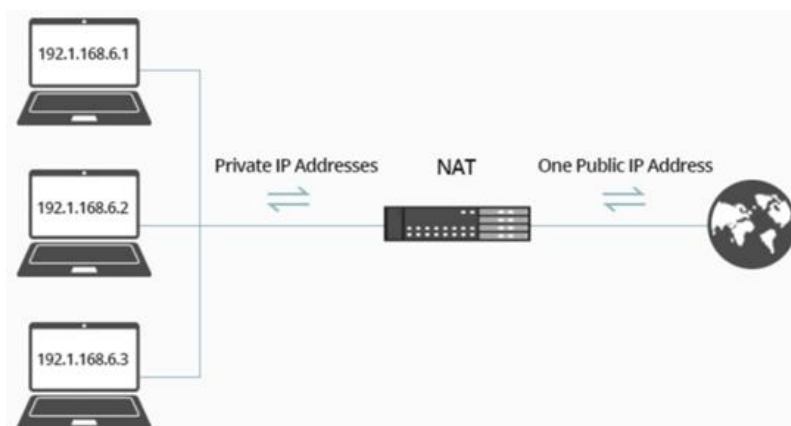


Figure 1-2: Overview of PAT. [1]

For every type of NAT, we need to create an object on the ASA. Also, ASA can act as a DHCP server, dynamically assigning IP addresses to internal devices. [1]

## 1.5 Access Control Lists (ACLs)

A collection of guidelines called an access control list (ACL) is used to regulate network traffic and improve security. An access control entry is what each rule in an ACL is known as. Even though the entire phrase is frequently used in official documentation, network experts frequently just call them access lists. To decide whether to forward or drop packets, ACLs filter network traffic at router interfaces. The router determines whether to allow or refuse a packet by comparing it to the ACL when it reaches an interface. Protocol type, port number/name, source and destination IP addresses, and other specified criteria can be used to permit/reject packets. [1] From all types of ACLs, we will use named extended ones, which can be applied as follows:

1. Create rules for denying /permitting your packet. Those include source, destination, protocol used, and even port numbers if we want to. [1]

```
ciscoasa(config)# access-list <acl-name> extended permit <protocol> <src-addr> <dst-addr>
```

2. Then, use the access-group command to bind the ACL to an interface in a specific direction:[1]

```
ciscoasa(config)# access-group <acl-name> <in |out> interface <interface-name>
```

In our experiment, ACLs will be used to allow flowing of traffic from lower to higher security level. (from outside to DMZ)

## 1.6 Interface related configuration

### 1.6.1 Configure Interfaces and Security Zones

We can change the name of an interface, the security-level and give IP address for it using the following commands:[1]

```
ASA-Birzeit(config)# interface <interface-name>  
ASA-Birzeit(config-if)# nameif <zone-name>  
ASA-Birzeit(config-if)# security-level <security-level>  
ASA-Birzeit(config-if)# ip address <ip-address> <subnet-mask>  
ASA-Birzeit(config-if)# no shutdown
```

### 1.6.2 Static Route Configuration

To make sure the ASA firewall is aware of how to forward traffic intended for external networks, we can define a default route or a route to a particular destination network using instructions as follows:[1]

```
ciscoasa (config)# route <interface-name> <dst-network> <subnet -mask> <next -hop-ip-address>
```

### 1.6.3 DHCP Configuration

In Cisco ASA, the DHCP server must be enabled on each interface where it's needed, allowing the firewall to automatically assign IP addresses and network settings to devices in that specific zone.

```
ciscoasa (config)# dhcpd address <start -ip> -< end-ip> <interface -name>  
ciscoasa (config)# dhcpd dns <dns-server -ip>  
ciscoasa (config)# dhcpd enable <interface-name>
```

it will give the default gateway automatically of the one associated with the interface that we run DHCP into it.

## 1.7 Dynamic Routing

In this lab, OSPF (Open Shortest Path First) was used for dynamic routing within the same autonomous system, which allows internal routers to exchange routing information efficiently. BGP (Border Gateway Protocol) was configured to enable communication between different autonomous systems-between Birzeit and Amazon networks. Route redistribution between OSPF and BGP ensured routing across both internal and external networks. [1] Configuration commands of them will be mentioned in the procedure since we aren't really interested of them at this point.

## 2 Procedure and Discussion

### 2.1 Setting up the Topology

#### 2.1.1 Topology Diagram & Components

The network was designed according to the topology illustrated in Figure 2-1, where the variable  $x$  represents the last two digits of my student ID (i.e., 82).

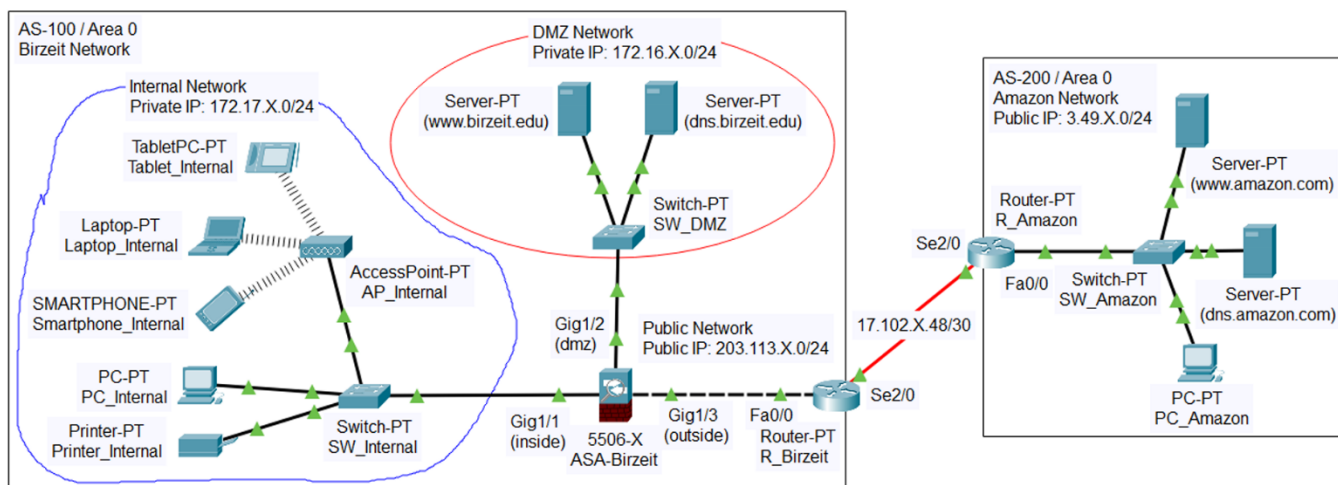


Figure 2-1: Topology. [1]

Table 2-1 shows the devices and connections implemented in the simulation:

Table 2-1: Used Devices.

| Network Devices    | End Devices      | Connections                                                      |
|--------------------|------------------|------------------------------------------------------------------|
| 1. Router-PT       | 1. PC-PT         | Automatic connections (start from the routers to avoid problems) |
| 2. Switch-PT       | 2. Server-PT     |                                                                  |
| 3. ASA 5506-X      | 3. Laptop-PT     |                                                                  |
| 4. Access Point-PT | 4. TabletPC-PT   |                                                                  |
|                    | 5. SMARTPHONE-PT |                                                                  |

## 2.1.2 IPs Distribution

The network topology in Figure 2-1 defined the first three octets of the IP addresses for all devices, while the fourth octet (host portion) is assigned according to the numbering convention specified in Figure 2-2.

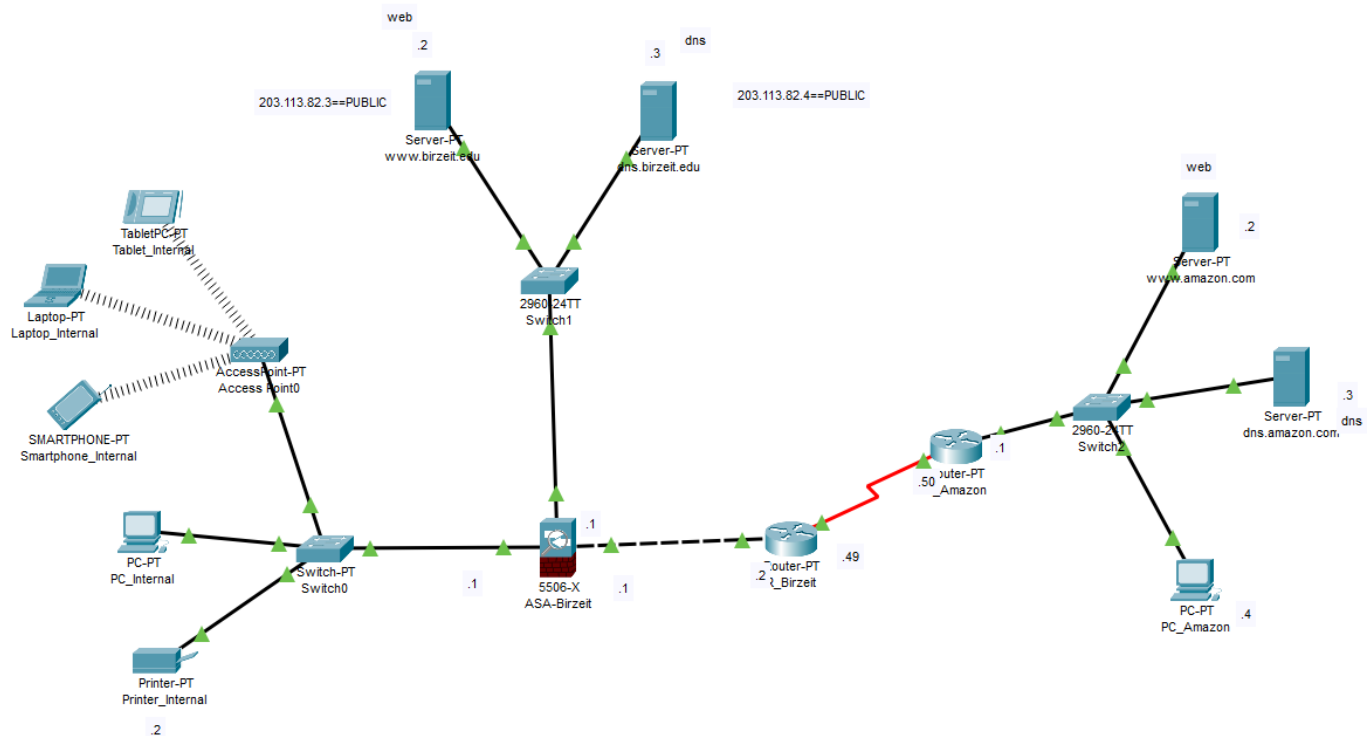


Figure 2-2: IP Distribution.

## 2.1.3 IP Configuration (All but ASA)

### 2.1.3.1 Routers Configuration

To give IP for a router, we can use the following commands

```
Router(config)#Interface <TYPE> <SLOT> / <PORT>
Router(config-if)#ip address <IP ADDRESS> <MASK>
Router(config-if)#no shutdown
```

For Birzeit router, it has 2 connected interfaces. To configure the one connected with the firewall (e.g. Fa 0/0) we can apply the following:

```
R_Birzeit(config)# interface FastEthernet0/0
```

```
R_Birzeit(config-if)# ip address 203.113.82.2 255.255.255.0
R_Birzeit(config-if)# no shutdown
R_Birzeit(config-if)# exit
```

To configure Se 2/0:

```
R_Birzeit(config)# interface Serial2/0
R_Birzeit(config-if)# ip address 17.102.82.49 255.255.255.252
R_Birzeit(config-if)# no shutdown
R_Birzeit(config-if)# exit
```

Figure 2-3 illustrates resulting modifications with an image for the router's interfaces.

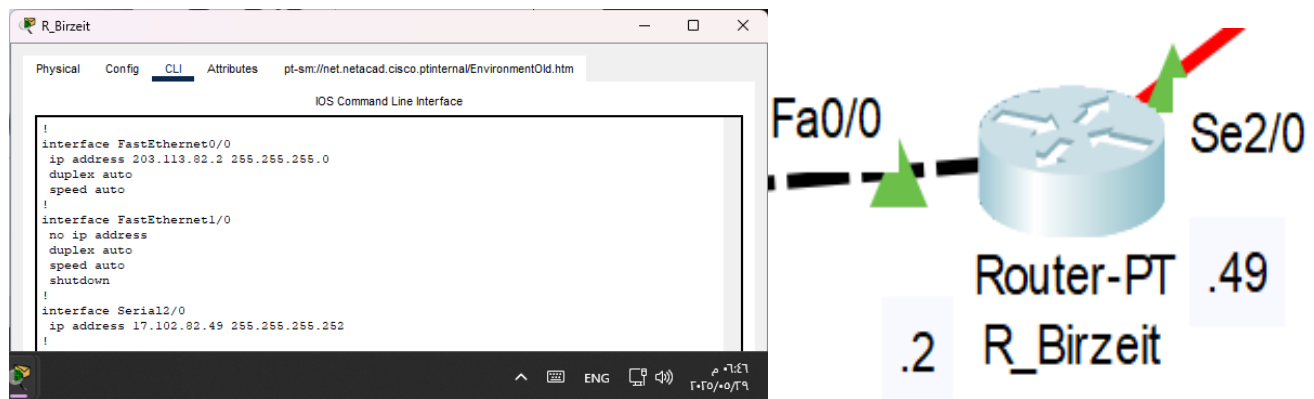


Figure 2-3: Interfaces modifications.

for R\_Amazon By CLI:

```
R_Amazon(config)#interface FastEthernet0/0
R_Amazon(config-if)#ip address 192.82.0.5 255.255.255.252
R_Amazon(config-if)#no shutdown
R_Amazon(config-if)#exit
R_Amazon(config)#interface Serial2/0
R_Amazon(config-if)#ip address 17.102.82.50 255.255.255.252
R_Amazon(config-if)#no shutdown
R_Amazon(config-if)#exit
```

Figure 2-4 illustrates resulting modifications with an image for the router's interfaces.

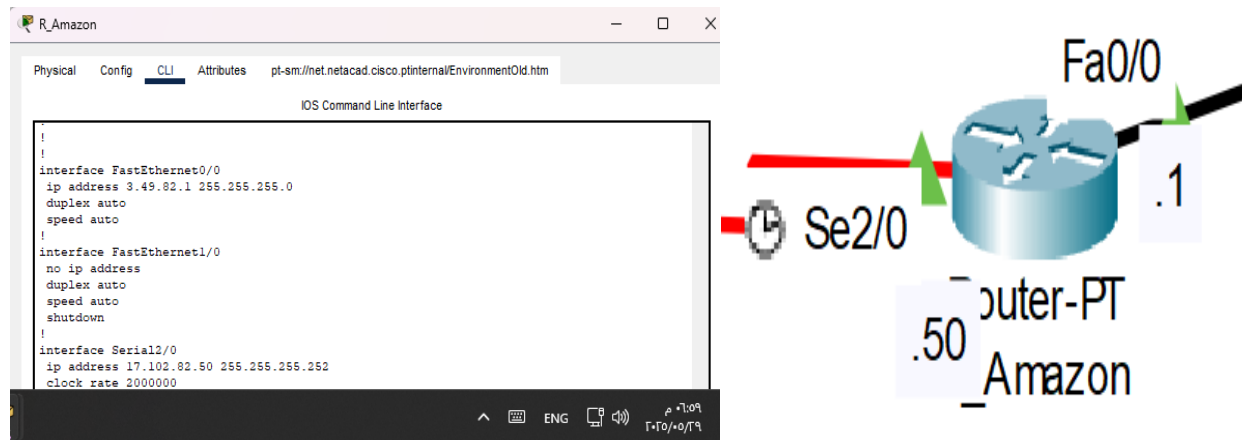


Figure 2-4: Interfaces modifications.

### 2.1.3.2 End Devices IP Configuration

#### 2.1.3.2.1 Personal Computer (PC), Personal Mobile Devices (PMD), and the Printer

In this experiment, we will use static IP configuration and Dynamic Host Configuration Protocol (DHCP), The Printer Internal and PC Amazon are configured with static IP addresses, including their default gateway and DNS server settings. Each device must have a unique IP address within the valid host range of its subnet.

Figure 2-5 shows How to configure IP addresses on PC Amazon.

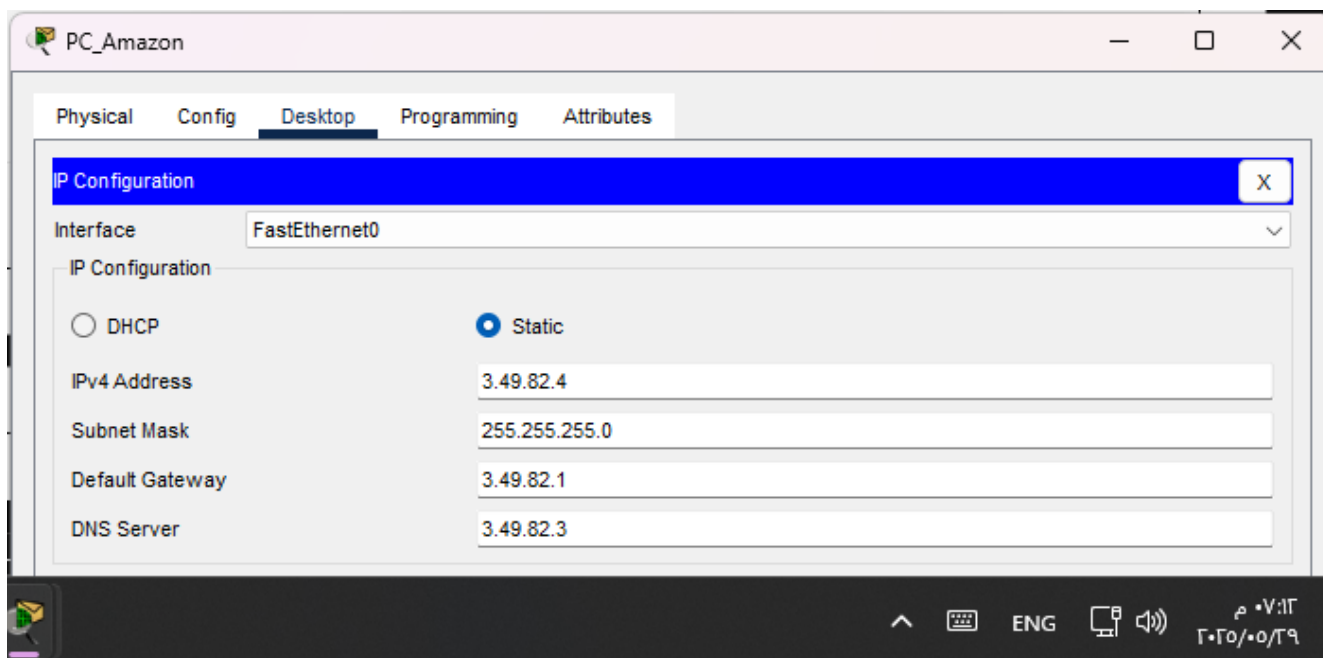


Figure 2-5: PC\_Amazon IP configuration.

Figure 2-6 shows IP configuration for the Printer\_Internal.

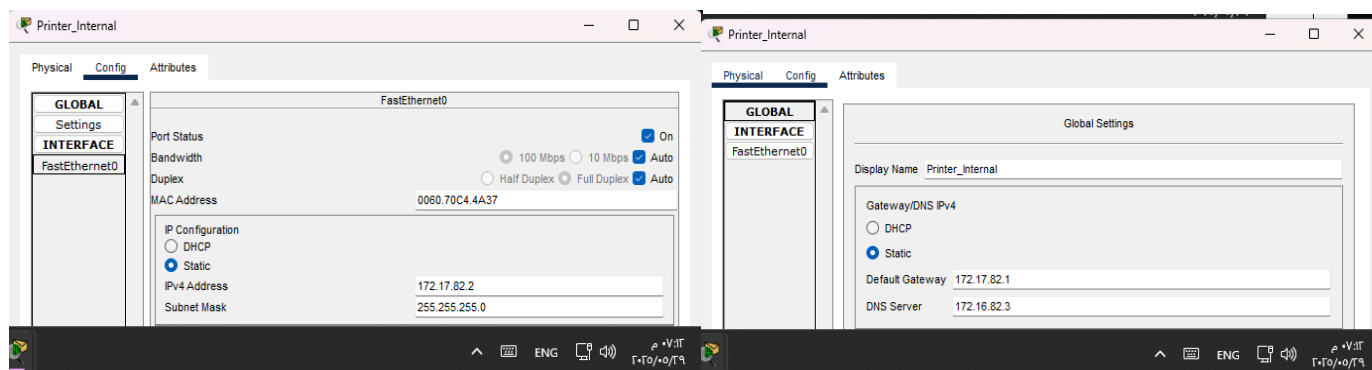
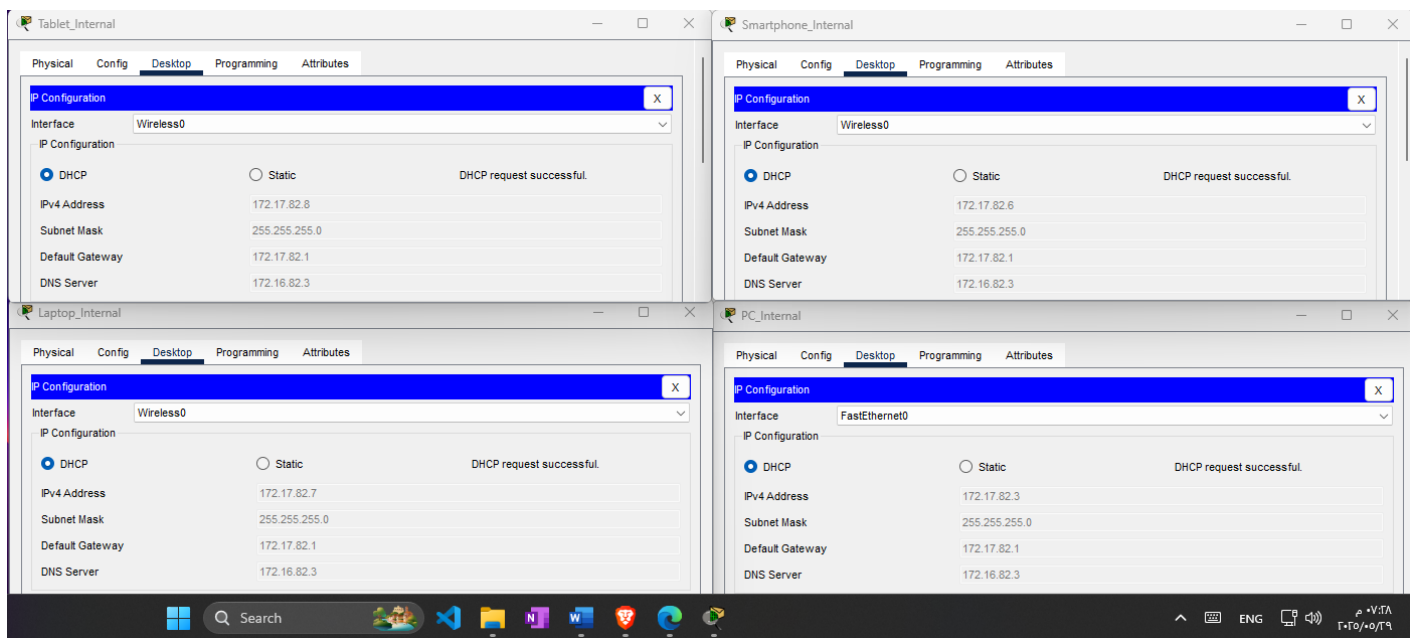


Figure 2-6: Printer\_Internal IP configuration.

Other Internal devices are configured using DHCP applying it in the firewall as follows:

```
ASA-Birzeit(config)# dhcpd address 172.17.82.3-172.17.82.102 inside
ASA-Birzeit(config)# dhcpd dns 172.16.82.3
ASA-Birzeit(config)# dhcpd enable inside
```

The result of latter is giving all internal devices an appropriate IP configuration according to Figure 2-7 . However, that wasn't straightforward, it turned out that there is a glitch in packet tracer, in which the ASA don't give addresses correctly. To review the problem, you can refer to page 33.



*Figure 2-7: Internal end devices IP configuration.*

#### 2.1.3.2.2 Servers

There are four servers in this topology: Two web servers and two DNS servers. Configuration of Birzeit servers are shown below in Figure 2-8

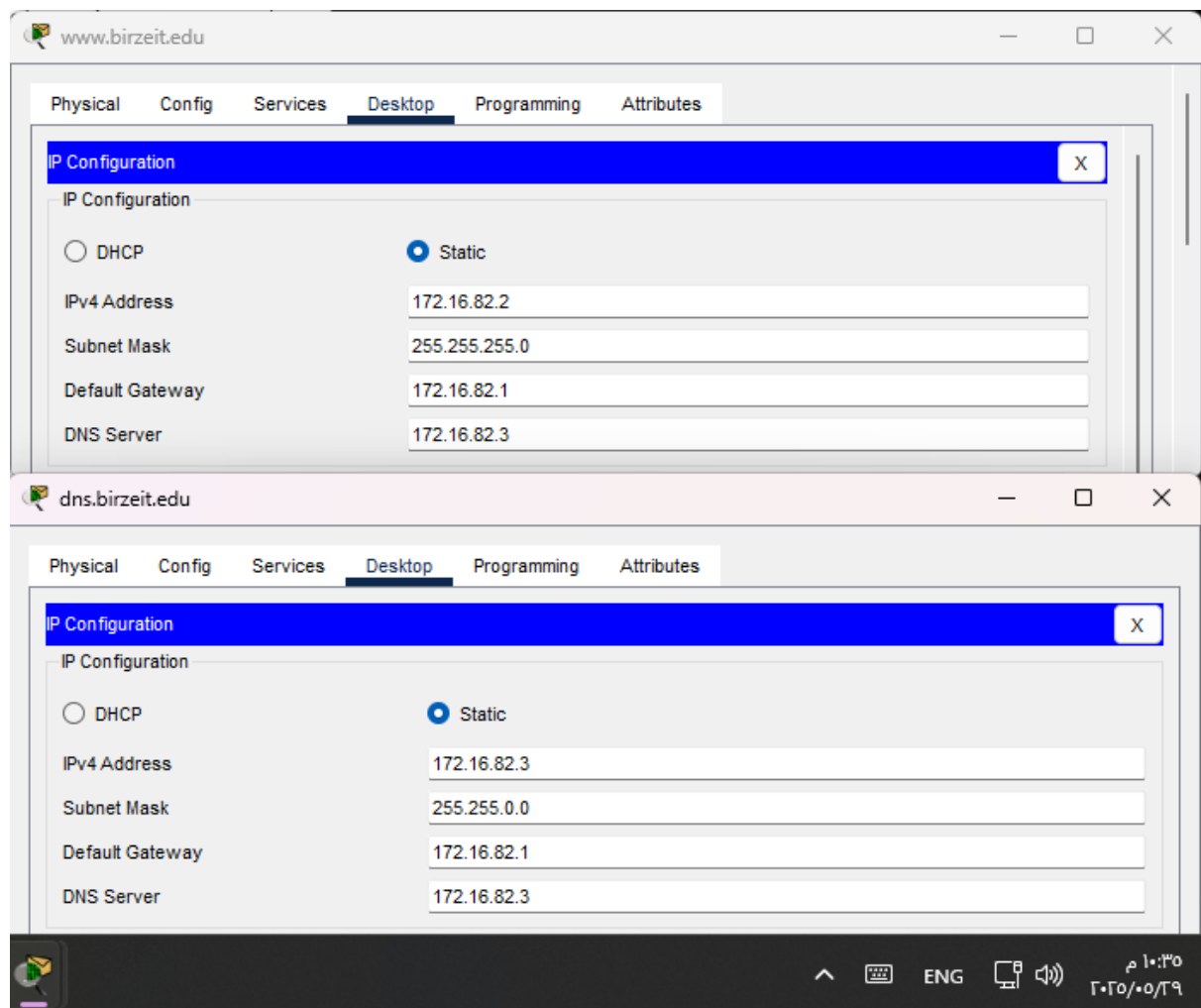


Figure 2-8: Birzeit servers IP configuration.

Configuration of Amazon servers are shown below in Figure 2-9

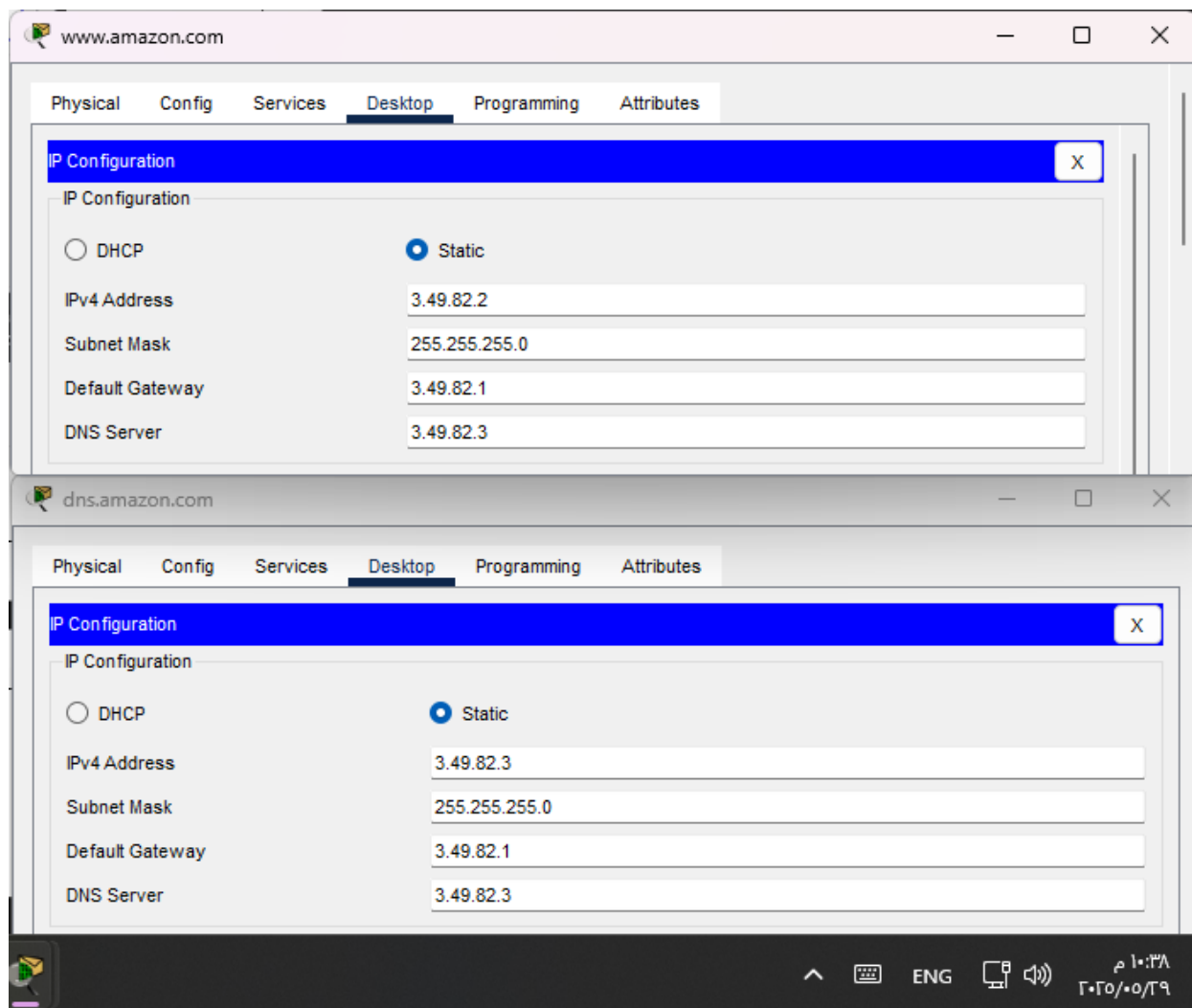


Figure 2-9: Amazon servers IP configuration.

#### 2.1.4 Servers' services configuration

For every web server we have disabled all the services and enable HTTP, and for DNS servers we have disabled all the services and enabled DNS. I let web server of Amazon view **Amazon E-commerce Company** . And for Birzeit: **Birzeit University** as follows in Figure 2-10

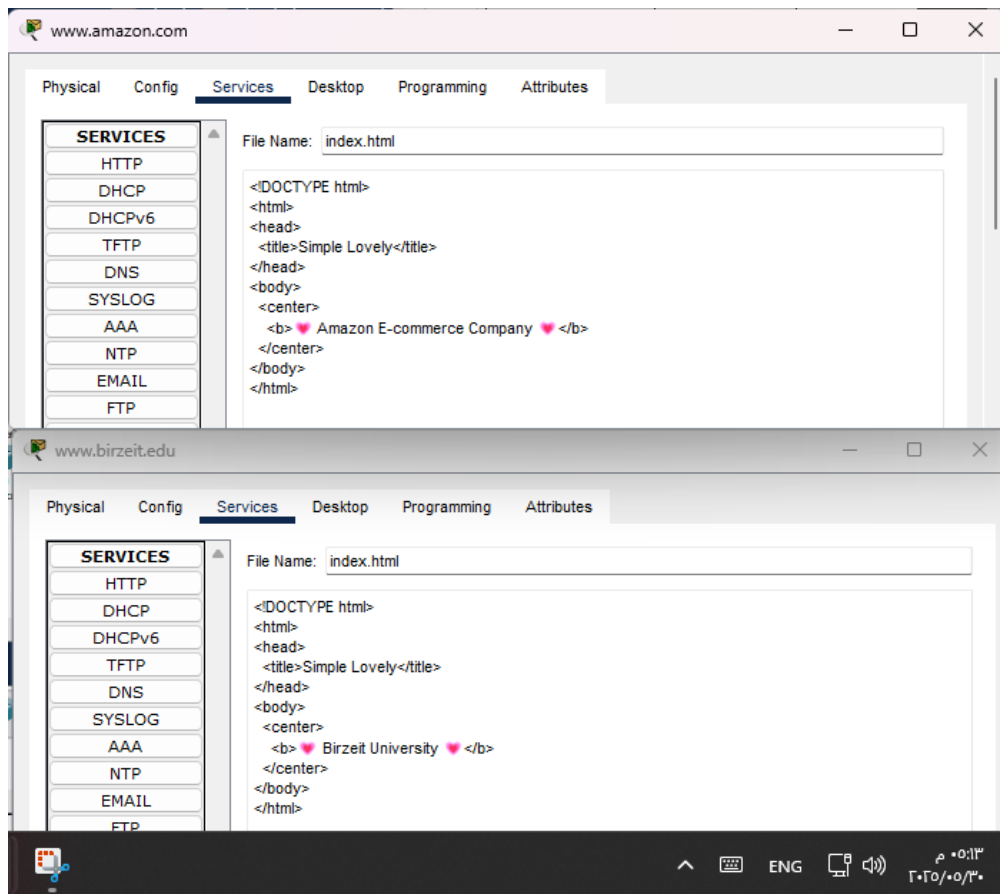


Figure 2-10: web servers main index configurations.

For Birzeit dns server I inserted the following Resource Records (RR) in Table 2-2.

Table 2-2: Birzeit DNS server RR.

| Name            | Value                               | Type |
|-----------------|-------------------------------------|------|
| www.birzeit.edu | 172.16.82.2 (IP of www.birzeit.edu) | A    |
| dns.amazon.com  | 3.49.82.3 (IP of dns.amazon.com)    | A    |
| amazon.com      | dns.amazon.com                      | NS   |

For Amazon dns server I inserted the following Resource Records (RR) as in Table 2-3.

Table 2-3: Amazon DNS server RR.

| Name            | Value                                               | Type |
|-----------------|-----------------------------------------------------|------|
| www.amazon.com  | 3.49.82.2 (IP of amazon.com)                        | A    |
| dns.birzeit.com | 203.113.82.4 ( <b>Public</b> IP of dns.birzeit.edu) | A    |
| birzeit.edu     | dns.birzeit.edu                                     | NS   |

Practically, that will be done using GUI shown in Figure 2-11 and Figure 2-12.

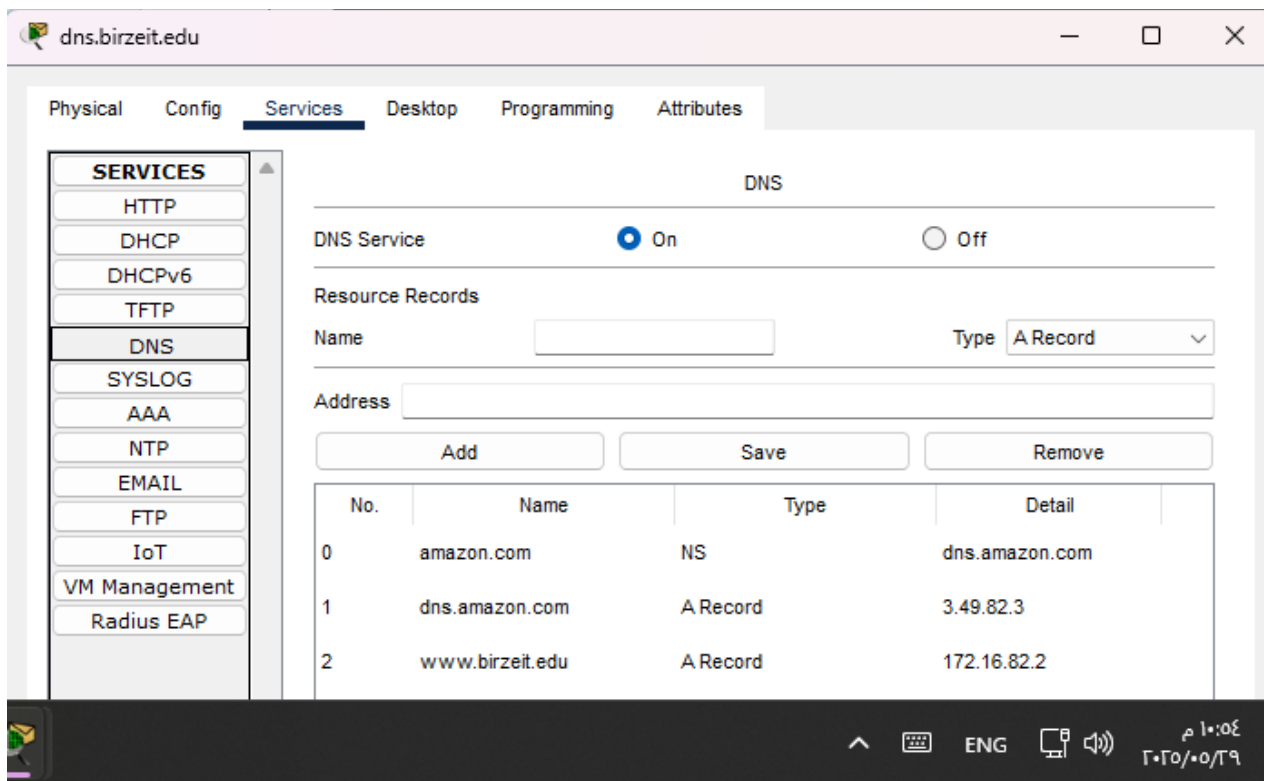


Figure 2-11: DNS records for Birzeit server.

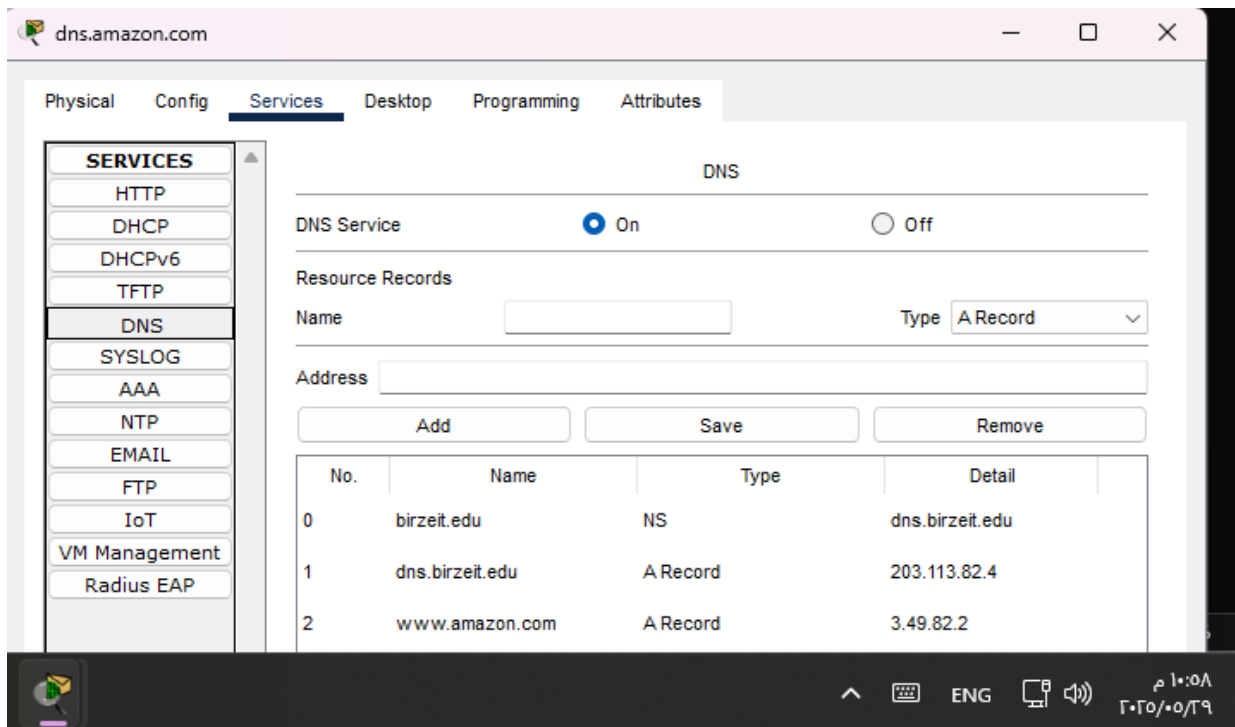


Figure 2-12: DNS records for Amazon server.

## 2.2 IGP (OSPF) Configuration redistributed with EGP ( BGP)

After IP configuration, we should configure OSPF and BGP for every router as follows:

- We will run the OSPF process on every router by commanding [1]  
`router ospf <PROCESS-ID>`
- We can use any number as a process-id, but we will use 1
- Now, every network connected to this router except anyone connected to different autonomous systems should be added to the OSPF process. To do that We can use the command [1]  
`network <NETWORK-ID> <WILDCARD-MASK> area <AREA-ID>`
- Configuring BGP is straightforward. For every router that resides between two Autonomous Systems (ASes), we need to enable BGP by specifying the AS number using the following command: `router bgp <AS-NUMBER>` where AS-NUMBER is for the router itself. [1]
- Inside the process of bgp we must establish peer relationships with neighboring routers in different ASes (eBGP) To do this, we use the command: `neighbor <NEIGHBOR-IP> remote-as <AS-NUMBER>` where AS-NUMBER is for the neighbor router.

- Table 2-4 outlines the methodology for configuring OSPF and BGP with redistribution between them routing across the network topology.

*Table 2-4: OSPF and BGP configuration.*

| Router Name      | Commands                                                                                                                                                                                                                                                                                                                                         |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R_Birzeit</b> | <pre> R_Birzeit(config)# router ospf 1 R_Birzeit(config-router)# network 203.113.82.0 0.0.0.255 area 0 R_Birzeit(config-router)# redistribute bgp 100 subnets R_Birzeit(config-router)#exit R_Birzeit(config)# router bgp 100 R_Birzeit(config-router)# neighbor 17.102.82.50 remote-as 200 R_Birzeit(config-router)# redistribute ospf 1 </pre> |
| <b>R_Amazon</b>  | <pre> R_Amazon(config)# router ospf 1 R_Amazon (config-router)# network 3.49.82.0 0.0.0.255 area 0 R_Amazon (config-router)# redistribute bgp 200 subnets R_Amazon (config-router)#exit R_Amazon (config)# router bgp 200 R_Amazon (config-router)# neighbor 17.102.82.49 remote-as 100 R_Amazon (config-router)# redistribute ospf 1 </pre>     |

## 2.3 Configuration of Cisco ASA Firewall

Configuring network interfaces, allocating security levels, establishing NAT (Network Address Translation), putting access control policies into place, and inspection enabling are all common steps in setting up a Cisco ASA firewall. DHCP configurations were discussed in on page 9 . Other common configuration for setting up a Cisco ASA firewall with a DMZ and NAT using standard commands will be discussed below.

### 2.3.1 Configuring Privileged Mode Password

A Cisco ASA firewall's privileged mode password is blank by default, which means it isn't configured. As a result, you can just hit Enter when asked for the privileged mode password. Use the following commands to change the password to "encs4130":

```
ASA-Birzeit> enable
Password:
ASA-Birzeit# configure terminal
ASA-Birzeit(config)# enable password encs4130
```

As a result, we can't now enter privileged mode without entering the password. It should be mentioned that when typing the password, it will be invisible, but if you wrote it correctly it will give you the access.

### 2.3.2 Configuring Interfaces and Security Zones

Every ASA interface needs to be set up with an IP address, security level, name, and enable. That can be done using the following:

```
ASA-Birzeit(config)# interface Gig1/1
ASA-Birzeit(config-if)# nameif outside
ASA-Birzeit(config-if)# security-level 0
ASA-Birzeit(config-if)# ip address 203.113.82.1 255.255.255.0
ASA-Birzeit(config-if)# no shutdown
ASA-Birzeit(config-if)# exit
ASA-Birzeit(config)# interface Gig1/2
ASA-Birzeit(config-if)# nameif inside
ASA-Birzeit(config-if)# security-level 100
ASA-Birzeit(config-if)# ip address 172.17.82.1 255.255.255.0
ASA-Birzeit(config-if)# no shutdown
ASA-Birzeit(config-if)# exit
ASA-Birzeit(config)# interface Gig1/3
ASA-Birzeit(config-if)# nameif dmz
ASA-Birzeit(config-if)# security-level 50
ASA-Birzeit(config-if)# ip address 172.16.82.1 255.255.255.0
ASA-Birzeit(config-if)# no shutdown
ASA-Birzeit(config-if)# exit
```

Latter configuration will be visible in `show running-config` as in Figure 2-13 with an image showing ASA interfaces.

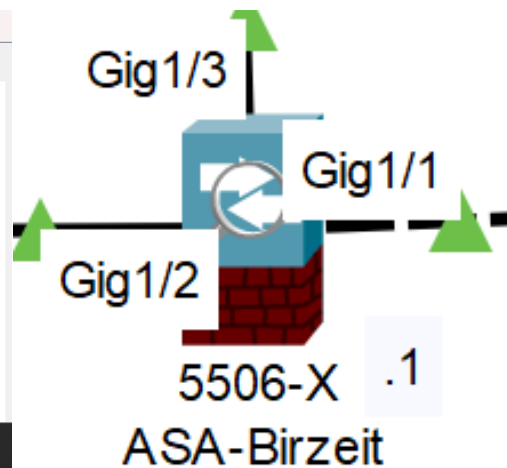
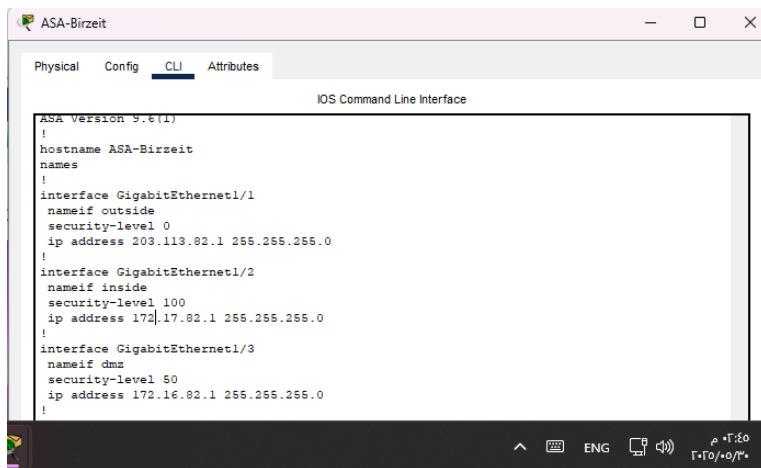


Figure 2-13: ASA interfaces configurations.

### 2.3.3 Network Address Translation (NAT) Configuration

For every type of NAT translation, we need to create what it is called an object; Objects in NAT can swap private addresses in outgoing packets to public IP addresses.

#### 2.3.3.1 Port Address Translation (PAT) configuration

We will configure interface ASA with PAT so that it can let devices in internal network use its public ip address using the following commands:

```

ASA-Birzeit(config)# object network INTERNAL-PAT
ASA-Birzeit(config-network-object)# subnet 172.17.82.0 255.255.255.0
ASA-Birzeit(config-network-object)# nat (inside,outside) dynamic interface

```

By that, every packet sent from internal network to outside will have its IP address being swapped with the public one that is assigned to interface Gig1/1 of the ASA

#### 2.3.3.2 Static NAT configuration

For the Web and DNS servers, we will use static NAT, which maps every private IP of them to a public IP according to Table 2-5 to make DMZ servers accessible from external networks

Table 2-5: Static NAT mappings

| Server                       | From (private IP) | To (public IP) |
|------------------------------|-------------------|----------------|
| Web server (www.birzeit.edu) | 172.16.82.2       | 203.113.82.3   |
| DNS server (dns.birzeit.edu) | 172.16.82.3       | 203.113.82.4   |

For the Web server we can type:

```
ASA-Birzeit(config)# object network DMZ-WEB-NAT
ASA-Birzeit(config-network-object)# host 172.16.82.2
ASA-Birzeit(config-network-object)# nat (dmz,outside) static 203.113.82.3
```

And for the DNS server we can use:

```
ASA-Birzeit(config)# object network DMZ-DNS-NAT
ASA-Birzeit(config-network-object)# host 172.16.82.3
ASA-Birzeit(config-network-object)# nat (dmz,outside) static 203.113.82.4
```

After finishing things up, configuration in Figure 2-14 have been shown

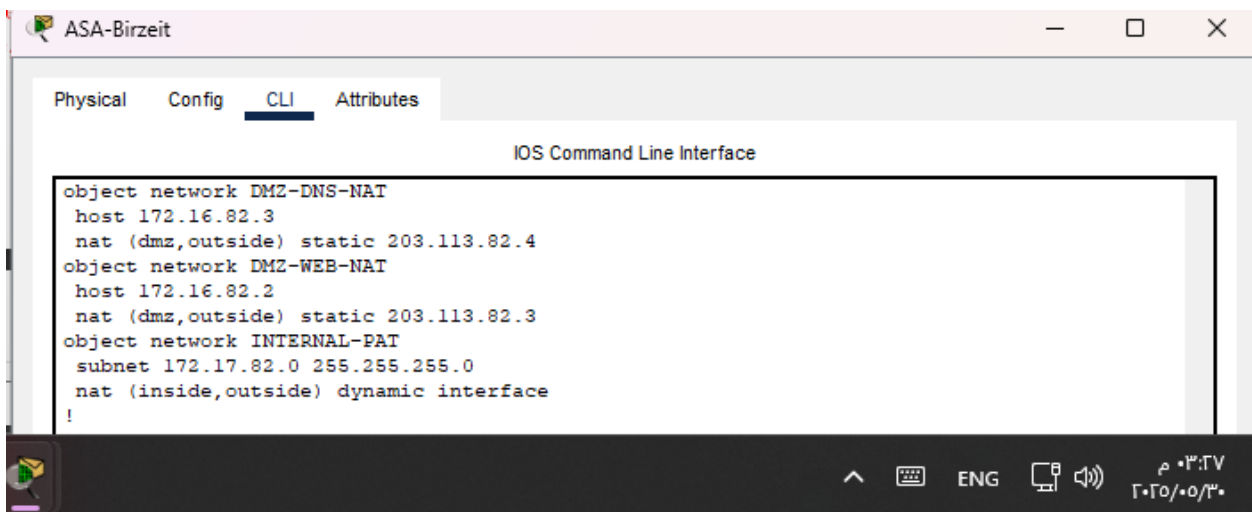


Figure 2-14: ASA zones settings.

### 2.3.4 Enabling External Access to DMZ Servers

to enable external users to use Web and DNS services hosted in the DMZ, we must make an Access control list (ACL) that allows the necessary traffic from external networks to the DMZ servers. After

that, we need to apply the ACL on the outside interface since it is the nearest to external network, with inbound configuration. That can be done using commands as follows:

```
ASA-Birzeit(config)# access-list OUTSIDE-ACCESS extended permit tcp any host 172.16.82.2 eq 80
ASA-Birzeit(config)# access-list OUTSIDE-ACCESS extended permit udp any host 172.16.82.3 eq 53
ASA-Birzeit(config)# access-group OUTSIDE-ACCESS in interface outside
```

By applying them, you should have the ability to reach DMZ section from outside by a browser or using `nslookup <hostname>` command

### 2.3.5 Enabling Return Traffic for HTTP, DNS, and ICMP to the Internal LAN

Because ASA firewall is stateful, it can monitor connections and permit protocol return traffic. This stops the blocking of genuine responses to internal clients from the DMZ or the Internet. DNS inspection is activated by default, but others aren't. we must explicitly enable HTTP and ICMP inspection by using the following commands which updates the global policy map:

```
ASA-Birzeit(config)# policy-map global_policy
ASA-Birzeit(config-pmap)# class inspection_default
ASA-Birzeit(config-pmap-c)# inspect http
ASA-Birzeit(config-pmap-c)# inspect icmp
```

After that we should check if everything is alright as in Figure 2-15



Figure 2-15: Inspection Enabling.

### 2.3.6 Configuring Default Routing

To enable Internet access for internal and DMZ devices, configure a default route on the ASA-Birzeit firewall that forwards all external traffic to the R Birzeit router using the ASA's outside interface. The destination 0.0.0.0 0.0.0.0 represents any IP address with any subnet mask, meaning that any traffic not matched by a more specific route will be directed to the next-hop IP address 203.113.82.2. The implementation of latter is as follows:

```
ASA-Birzeit(config)# route outside 0.0.0.0 0.0.0.0 203.113.82.2
```

## 2.4 Access Point Configuration and Related Network Settings

### 2.4.1 Access Point Configuration

The access point enables wireless connectivity within the internal network. To set up the wireless network, navigate to the “Config” tab and configure the access point with the following parameters: Set the network name to *BZU-WiFi*, use *WPA2-PSK* as the security mode, specify *EnCs4130* as the pre-shared key (which wireless devices must enter to connect), and select *AES* as the encryption type. Figure 2-16 illustrates how that is done in practice

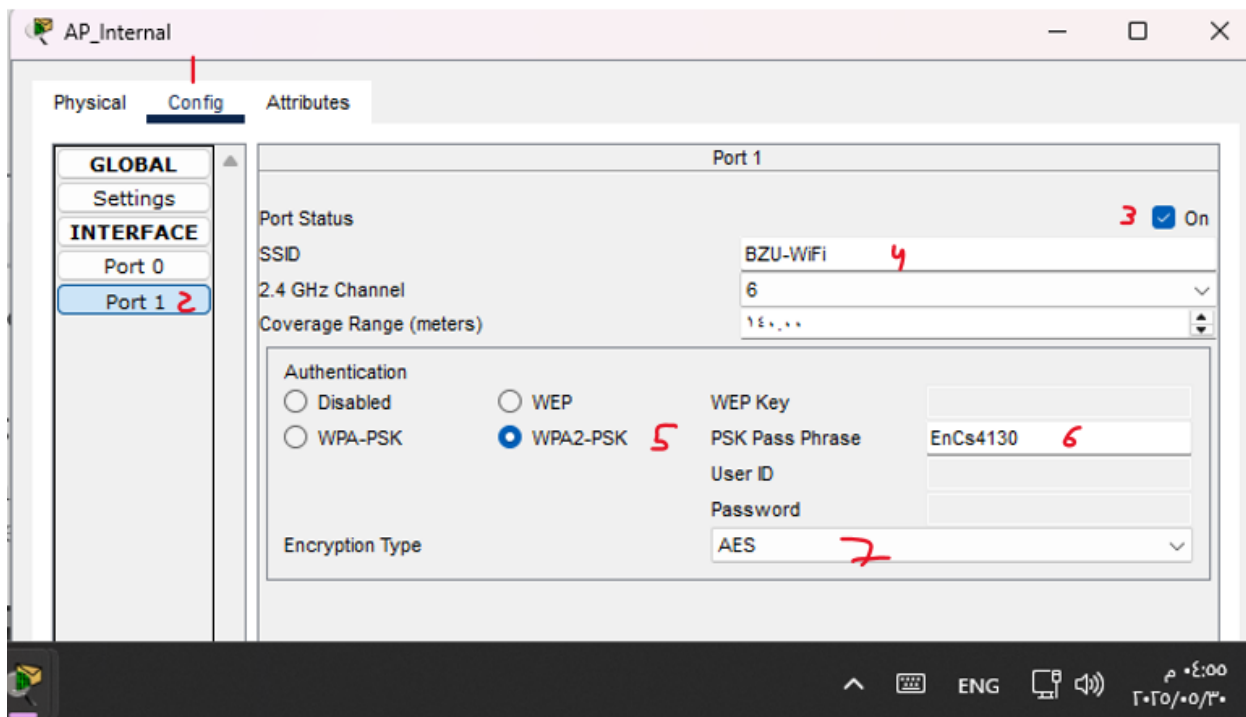


Figure 2-16: Access point modifications.

### 2.4.2 Connecting Wireless Devices

The tablet, laptop, and smartphone establish wireless connections to the internal network without any problems, but the laptop needs a WPC300N wireless adapter to enable wireless connectivity because it does not have an integrated wireless interface like the tablet and smartphone have. Click on the laptop, select the "Physical" tab, turn off the computer, take out the current FastEthernet module, add the WPC300N adapter from the list of modules, and then turn the laptop back on to install the adapter. After

installing the adapter, the laptop can connect to the BZU-WiFi network and be set up to dynamically obtain an IP address, as illustrated in Figure 2-17.

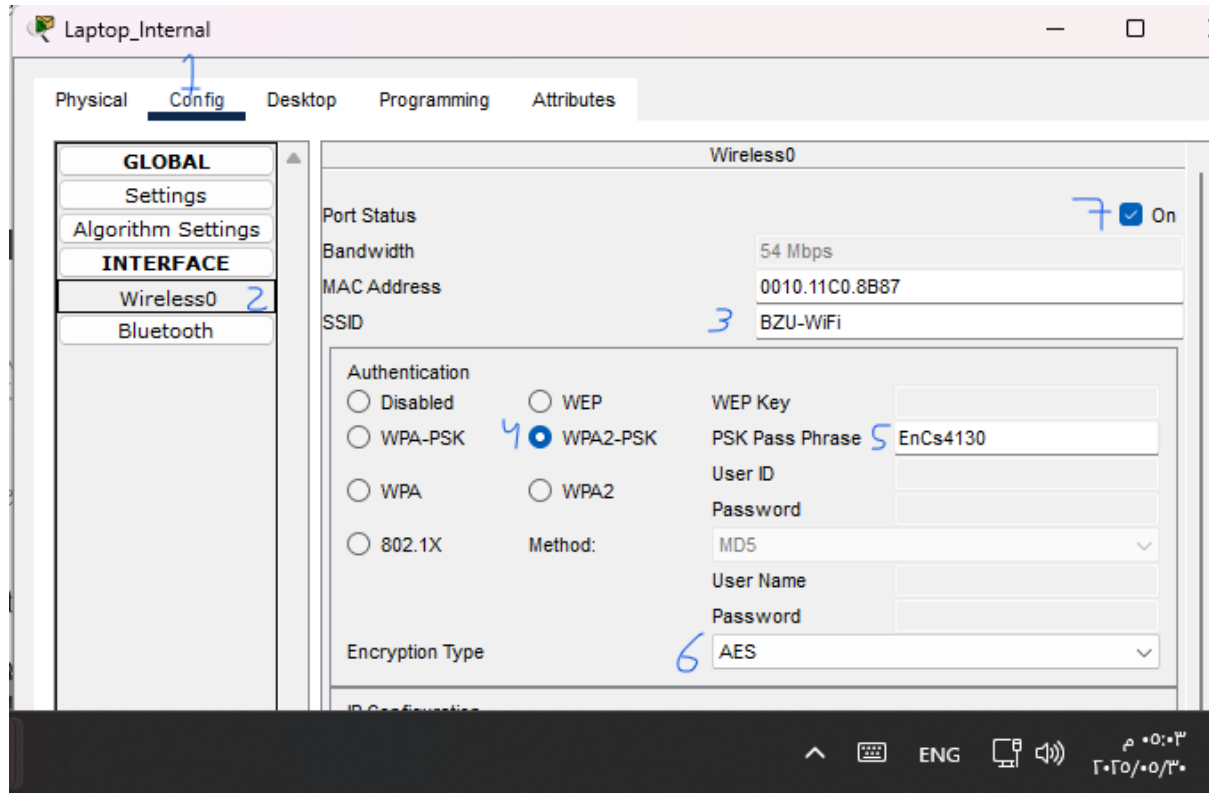


Figure 2-17: Connecting the Laptop with the Access point.

The tablet and smartphone can connect to the BZU-WiFi network in the same way as the laptop.

## 3 Results

### 3.1 Connectivity Between Devices

#### 3.1.1 Test 1– Accessing “www.birzeit.edu” from PC Amazon (Internet)

I tried to access Birzeit Web server from PC Amazon, and it worked. Using simulation mode, the packet has taken the following path:

(start)PC\_Amazon→ dns.amazon.com(to give the PC the address of web server using the Resource Records (RR) which are of type NS)→ PC\_Amazon→ R\_Amazon→ R\_Birzeit→ ASA\_Birzeit→ www.birzei.edu → ASA\_Birzeit→ R\_Birzeit→ R\_Amazon→ PC\_Amazon(end). Click [here](#) to see a video of packet simulation. After packet reception to PC, it viewed the result shown in Figure 3-1

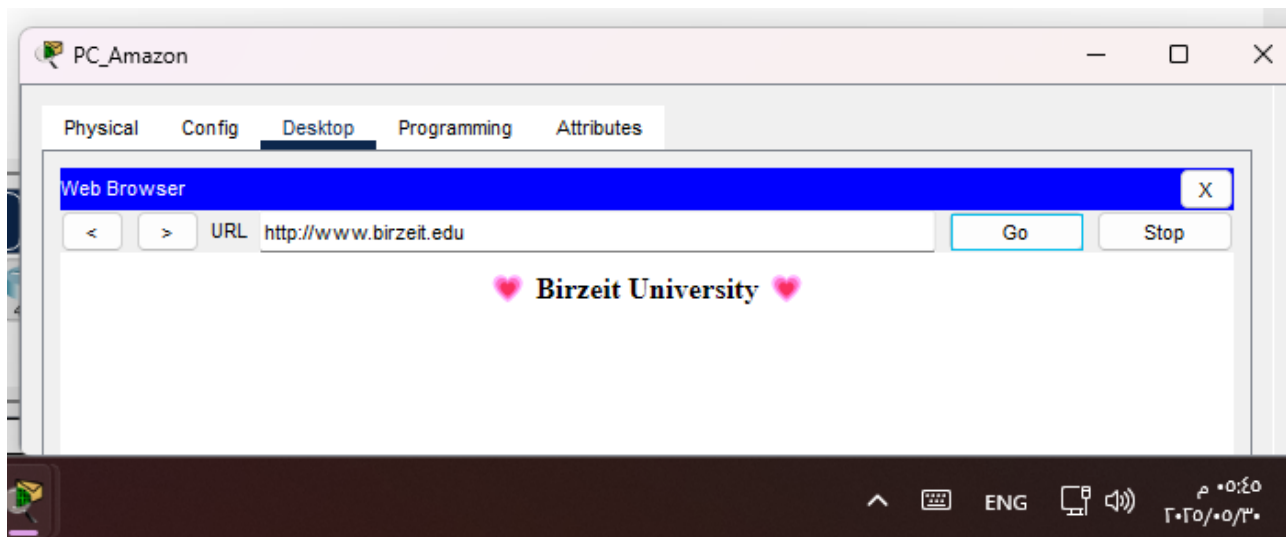


Figure 3-1: Accessing “www.birzeit.edu” from PC Amazon.

### 3.1.2 Test 2– Accessing “www.amazon.com” from Smartphone (Birzeit Internal Network)

We have could access amazon successfully as shown in Figure 3-2

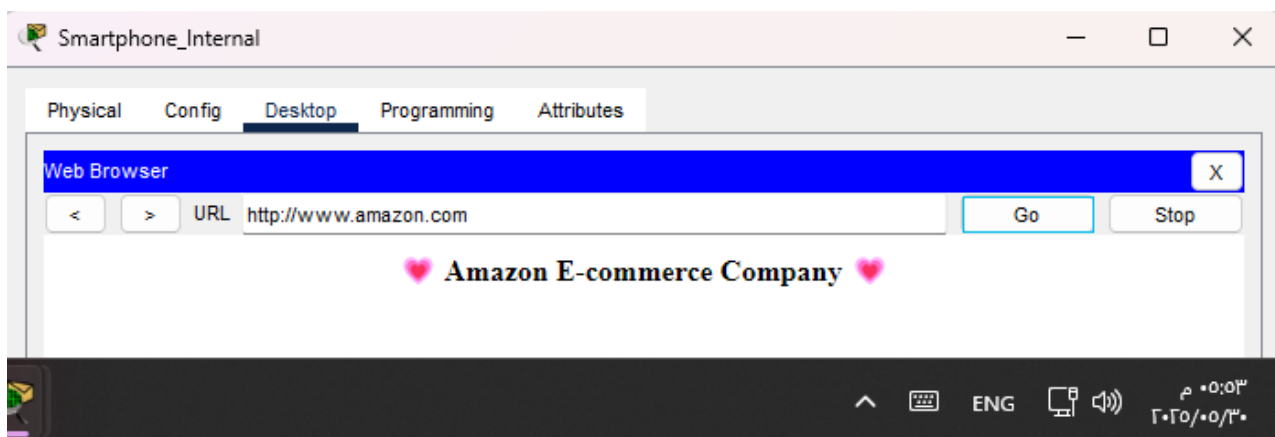


Figure 3-2: Accessing “www.amazon.com” from Smartphone.

The taken route from Smartphone\_Internal to Amazon was as follows( A video record for it is [here](#)):

(start)Smartphone\_Internal→ASA\_Birzeit→ dns.birzeit.edu(to give the PC the address of web server using the Resource Records (RR) which are of type NS) → ASA\_Birzeit→ Smartphone\_Internal → ASA\_Birzeit → R\_Birzeit → R\_Amazon → www.amazon.com → R\_Amazon→ R\_Birzeit→ ASA\_Birzeit→ Smartphone\_Internal(end).

### 3.1.3 Test 3– Pinging a Host in Birzeit Network from PC Amazon (Internet):

As seen in Figure 3-3, an attempt to send ICMP packets (ping) from the Amazon PC to a device connected to the Birzeit network would fail. Unless specifically allowed, firewall's default security policy prevents traffic from being sent from a lower security level-like the Internet-to a higher security level-like the internal network.

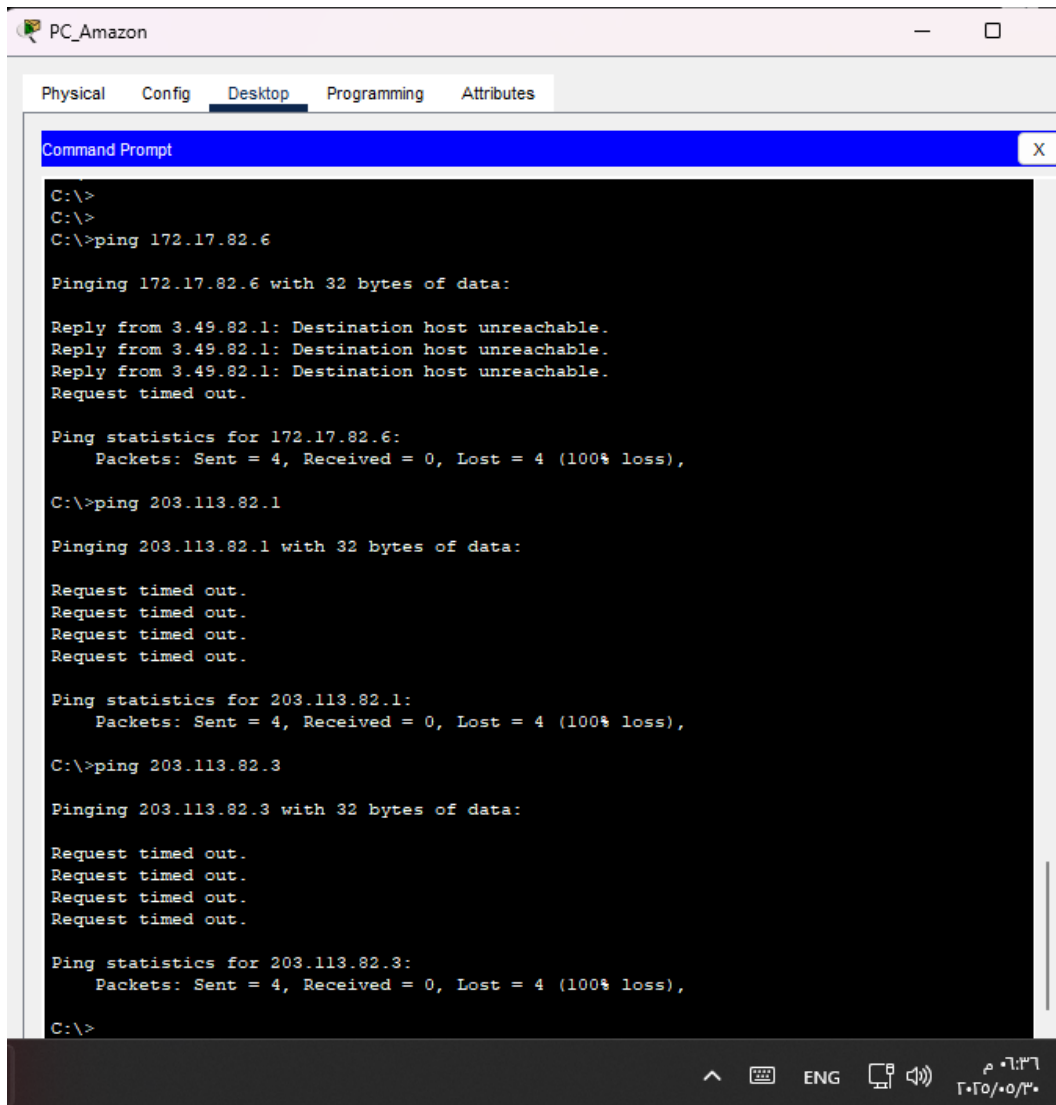


Figure 3-3: ICMP packets from Amazon PC to a device connected to Birzeit.

### 3.1.4 Test 4– Pinging PC Amazon (Internet) from a Host in Birzeit Network

The ping was successful as shown in Figure 3-4

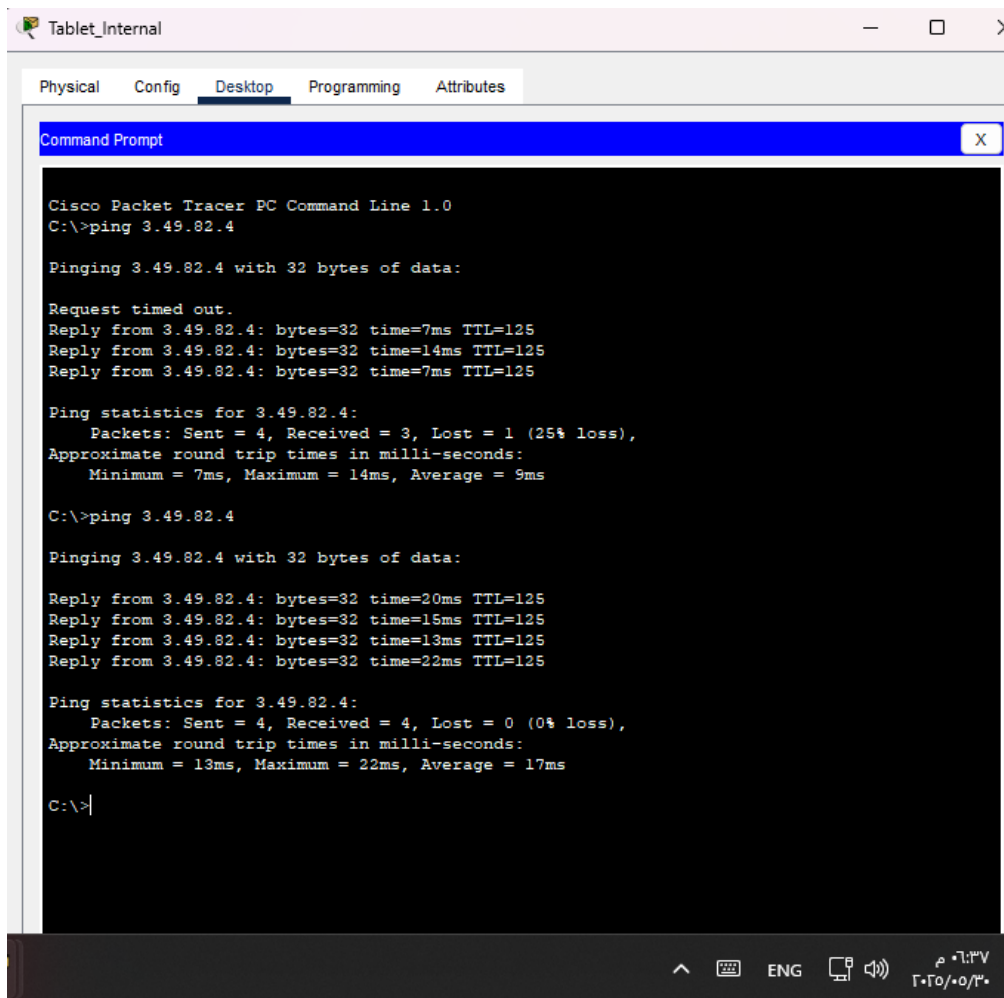


Figure 3-4: Pinging PC Amazon from a Host in Birzeit.

We have transmitted ICMP packets (ping) to the Amazon PC from the tablet connected to the Birzeit internal network. Because the ASA allows outgoing connections from higher to lower security levels, this traffic is permitted by default. Ping responses from the Internet (lower security level) back to the internal network (higher security level) are permitted as the ASA has ICMP inspection enabled. The latter match the Theory.

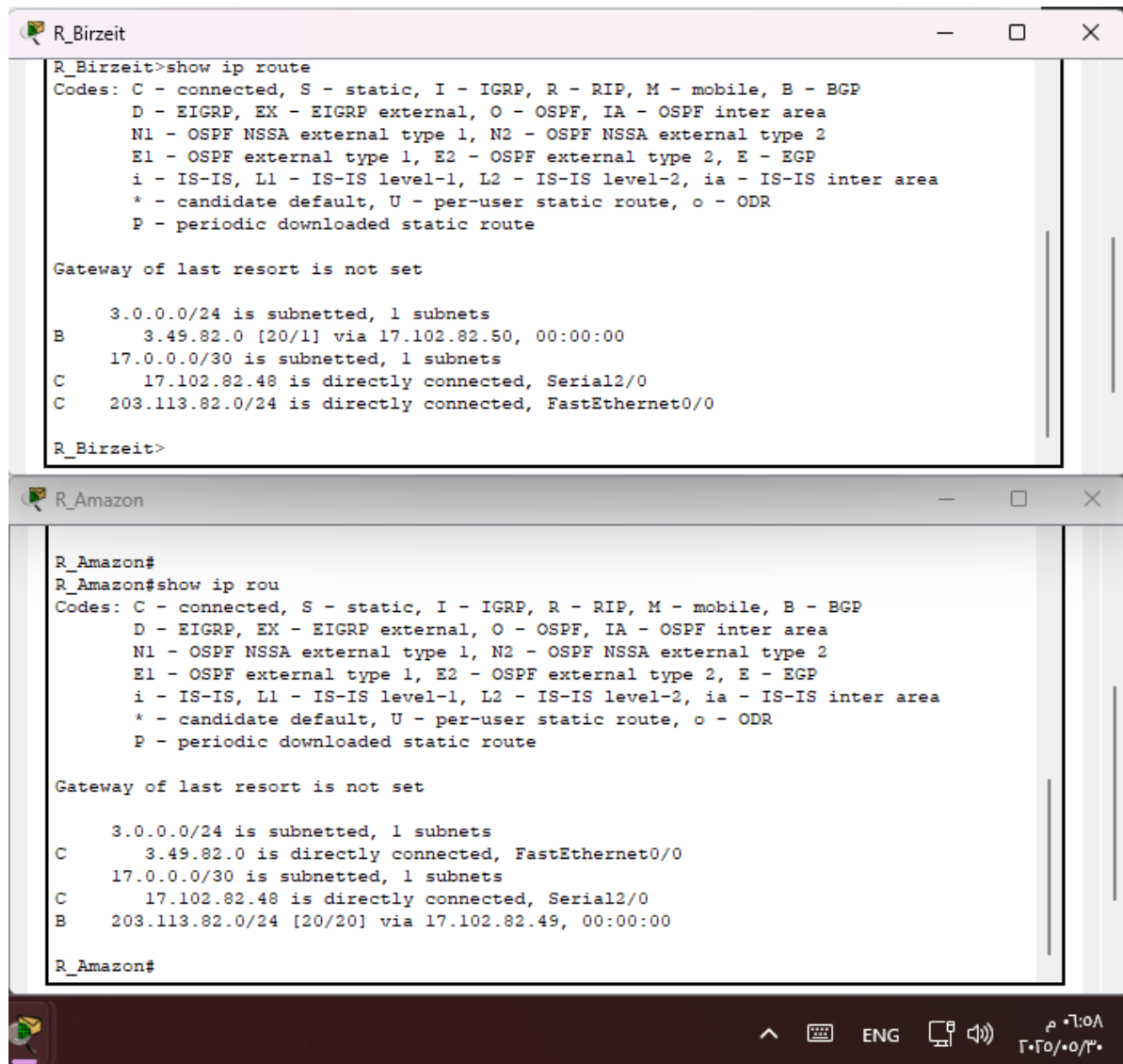
## 3.2 Configuration Verification

### 3.2.1 Routing Configuration

We can inspect the Routing configuration and monitor its behavior using the command:

```
Show ip route
```

Figure 3-5 show the application for that command:



```
R_Birzeit>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  3.0.0.0/24 is subnetted, 1 subnets
B       3.49.82.0 [20/1] via 17.102.82.50, 00:00:00
  17.0.0.0/30 is subnetted, 1 subnets
C       17.102.82.48 is directly connected, Serial12/0
C       203.113.82.0/24 is directly connected, FastEthernet0/0

R_Birzeit>

R_Amazon#
R_Amazon#show ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  3.0.0.0/24 is subnetted, 1 subnets
C       3.49.82.0 is directly connected, FastEthernet0/0
  17.0.0.0/30 is subnetted, 1 subnets
C       17.102.82.48 is directly connected, Serial12/0
B       203.113.82.0/24 [20/20] via 17.102.82.49, 00:00:00

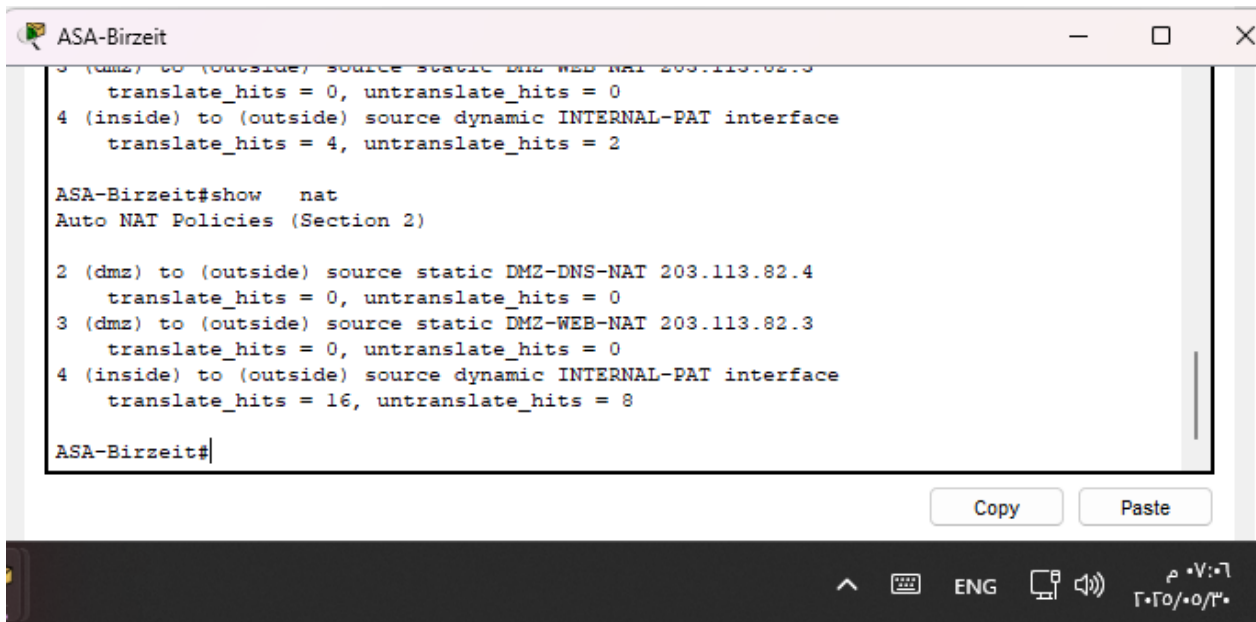
R_Amazon#
```

Figure 3-5: ASA Routing configurations.

The brackets containing two numbers on the form [a/b] refer to the [administrative distance/cost(metric) of routing protocol] respectively. This command provides information about which routes are reachable, how they are reached, through which interface and next-hop router, how the route was discovered, and the associated cost. From Figure 3-5 we see that network of ASA is known for both: one is directly connected to it and the other using BGP. Also, Amazon network is known in each router.

### 3.2.2 NAT Rules

We can review NAT rules using command `show nat` as shown in Figure 3-6



```
ASA-Birzeit#show nat
Auto NAT Policies (Section 2)

2 (dmz) to (outside) source static DMZ-DNS-NAT 203.113.82.4
  translate_hits = 0, untranslate_hits = 0
3 (dmz) to (outside) source static DMZ-WEB-NAT 203.113.82.3
  translate_hits = 0, untranslate_hits = 0
4 (inside) to (outside) source dynamic INTERNAL-PAT interface
  translate_hits = 16, untranslate_hits = 8

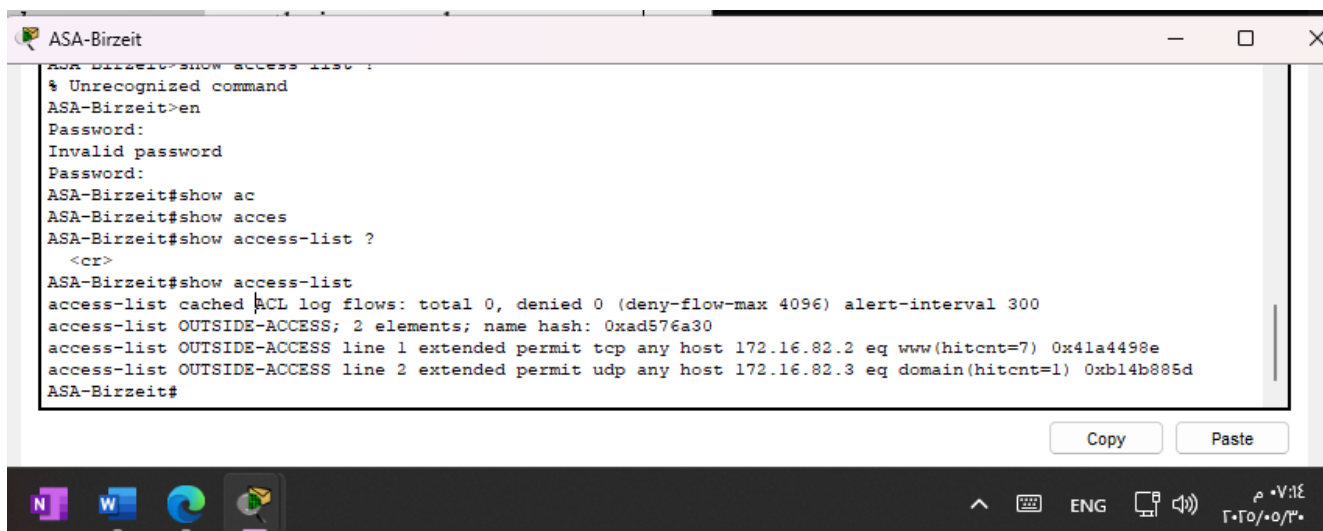
ASA-Birzeit#
```

Figure 3-6: NAT configurations.

Our theory has matched our result. The outcomes verify that the ASA firewall's NAT setup is operating as intended. The firewall's stateful inspection and NAT rules are effectively permitting and controlling outward and return traffic, as evidenced by the successful reception of answers from external servers and the ability of internal devices to access the Internet via PAT. The DMZ servers' static NAT configuration is accurate. All things considered, the firewall's behavior is consistent with our theoretical predictions, enabling safe and effective communication between internal, DMZ, and external networks.

### 3.2.3 ACL Rules

One can see ACL rules using the command `show access-list` as in Figure 3-7 which are the same rules we have configured in page 20

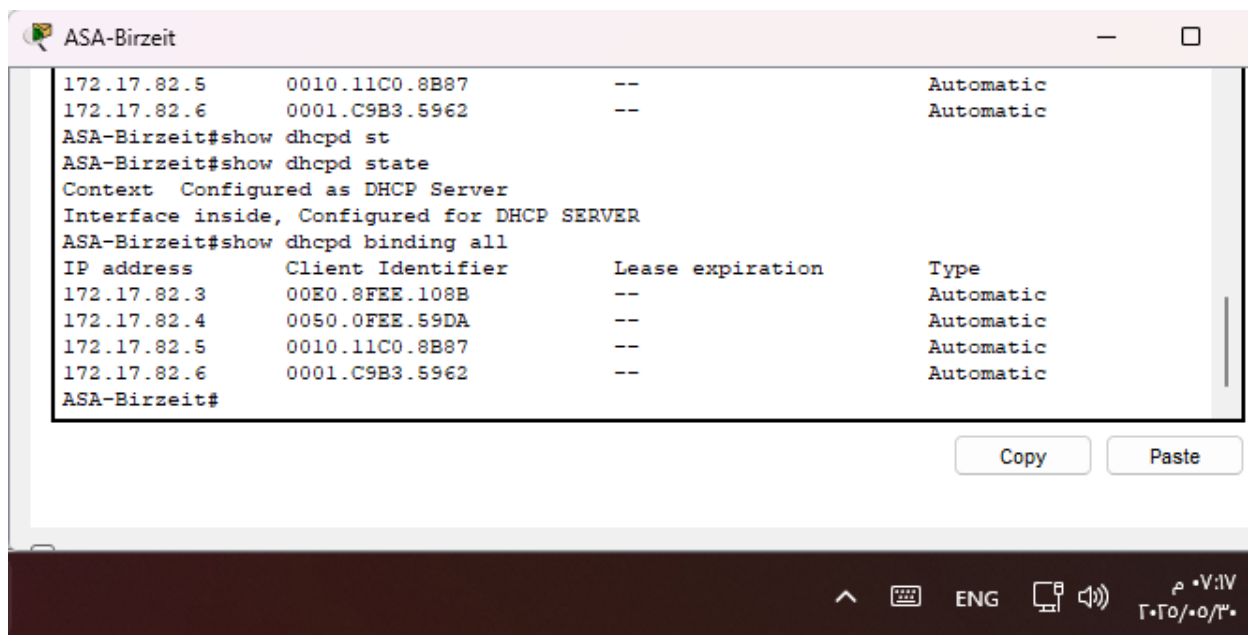


```
ASA-Birzeit>show access-list
% Unrecognized command
ASA-Birzeit>en
Password:
Invalid password
Password:
ASA-Birzeit#show ac
ASA-Birzeit#show acces
ASA-Birzeit#show access-list ?
<cr>
ASA-Birzeit#show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096) alert-interval 300
access-list OUTSIDE-ACCESS; 2 elements; name hash: 0xad576a30
access-list OUTSIDE-ACCESS line 1 extended permit tcp any host 172.16.82.2 eq www(hitcnt=7) 0x41a4498e
access-list OUTSIDE-ACCESS line 2 extended permit udp any host 172.16.82.3 eq domain(hitcnt=1) 0xb14b885d
ASA-Birzeit#
```

Figure 3-7: ACL Rules.

### 3.2.4 DHCP Leases && Wireless Connectivity for access points members

Using command like `show dhcpd state` and `show dhcpd binding all` we can review current DHCP Leases as in Figure 3-8



```
ASA-Birzeit
172.17.82.5      0010.11C0.8B87      --      Automatic
172.17.82.6      0001.C9B3.5962      --      Automatic
ASA-Birzeit#show dhcpd st
ASA-Birzeit#show dhcpd state
Context Configured as DHCP Server
Interface inside, Configured for DHCP SERVER
ASA-Birzeit#show dhcpd binding all
IP address      Client Identifier      Lease expiration      Type
172.17.82.3      00E0.8FEE.108B      --      Automatic
172.17.82.4      0050.0FEE.59DA      --      Automatic
172.17.82.5      0010.11C0.8B87      --      Automatic
172.17.82.6      0001.C9B3.5962      --      Automatic
ASA-Birzeit#
```

Figure 3-8: DHCP: Leases && interfaces.

It leased 4 addresses which matches with the number of devices in internal network that rely on DHCP. We see that the configuration is correctly assigned to interface *inside*. In Figure 3-9 PC\_Internal got its configuration using DHCP.

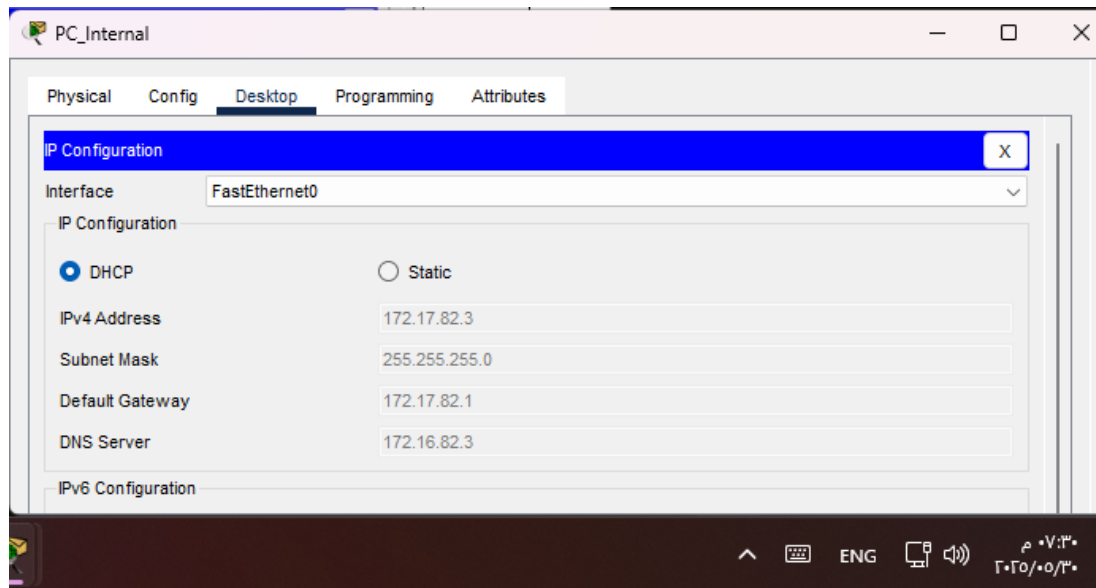


Figure 3-9: Using DHCP to give PC\_Internal IP configuration.

As shown in Figure 3-10 all devices have established successful connection to the private network and got IP configuration which matches the theory.

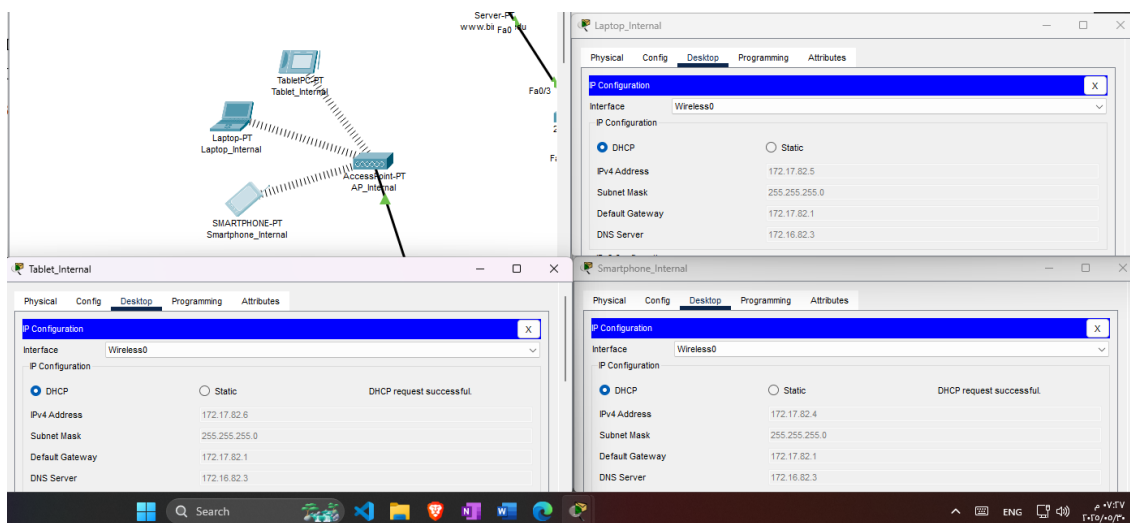


Figure 3-10: Using DHCP to give devices connected to the AP IP configuration .

## 4 Conclusion

This experiment showed the implementation of a secure network infrastructure using Cisco ASA Firewall and dynamic routing protocols. The main objective is to ensure secure communication between internal, DMZ, and external networks. It was successfully accomplished through the configuration of interfaces, security zones, NAT, ACLs, and DHCP. The use of Cisco Packet Tracer and diagnostic tools such as `ping`, `show ip route`, and `show access-list` strengthened my understanding of how a packet will flow, firewall behavior, and policy applying. Through this lab, I strengthened my knowledge in firewall configuration, network design, and the use of stateful inspection and access control lists in securing enterprises. Furthermore, I realized the importance of accurate planning, validation, and troubleshooting in maintaining robust network security and functionality.

## 5 References

- [1] C. Judd, "Network Security Zones," 9 October 2018. [Online]. Available: <https://www.kwtrain.com/blog/network-security-zones>. [Accessed 30 May 2025].
- [2] ENCS4130 Manual, Birzeit University, 2025.

## 6 Feedback

I feel this experiment was a highly valuable learning experience. Yes, it has combined important concepts such as ASA firewall configuration- NAT, ACLs, DHCP, and dynamic routing with OSPF and BGP, but the experiment was time-intensive and would benefit from being split into two sessions, possibly integrating Experiment No. 11. Additionally, the DHCP feature in Packet Tracer presented issues with DNS configuration in which it doesn't give correct DNS address for internal devices but give correctly for other fields like leased IP address, subnet mask, and default gateway. This should be addressed or clarified in future sessions or even use static assigning instead of DHCP. To overcome the problem in the .pkt file, you can just *reload* the ASA. Overall, the lab enhanced my understanding of secure network design and protocol implementation. Figure 6-1 shows how the problem looks like

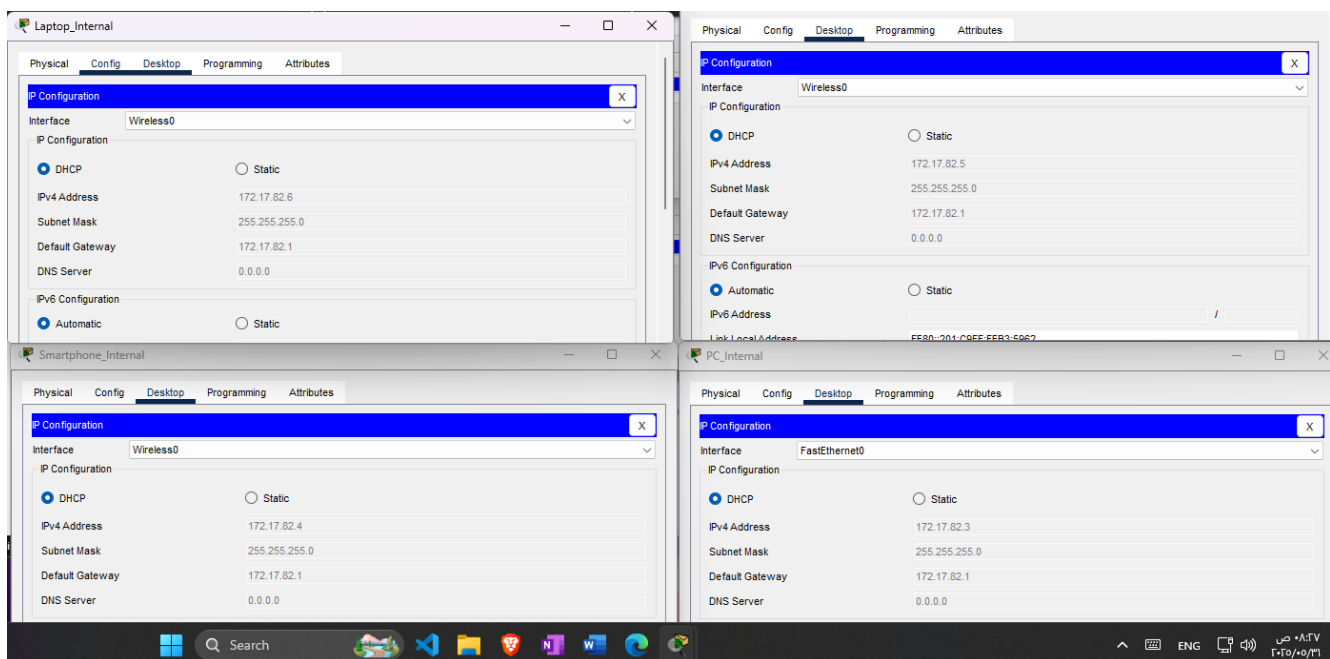


Figure 6-1: DNS setting failure.